"ResNet-18 and DenseNet-121 Architectures for OCT Retinal Analysis: Effects of Transfer Learning, Data Augmentation, Fine-Tuning Strategies and Model Calibration"

Abstract

Optical Coherence Tomography (OCT) retinal image analysis has become a critical component in automated diagnosis, using deep learning to detect subtle pathological features that are labor intensive for manual review. This study aimed to train a lightweight classifier without compromising performance, while investigating the interpretability of the results. For this task we selected ResNet18 and DenseNet121, as their architectures effectively mitigate vanishing gradients and promote feature reuse. The models were trained using transfer learning with gradual unfreezing of backbone layers on both augmented (Aug) and non-augmented (base) datasets. Based on validation performance, DenseNet121-Aug was selected for targeted fine-tuning on hard samples to further boost performance for cases which the model was overconfidently (> 0.9) predicting falsely. Hard-refinement led to a shift in Precision (+3.6%) for Drusen cases and Recall (+0.96%) for CNV cases, resulting in a final macro test F1-score of 95.01%. Finally, Grad-CAM and Occlusion Sensitivity was utilized to analyze the model's decision boundaries, confirming its effectiveness for reliable clinical decision-support systems.

1. Introduction

Retinal diseases such as choroidal neovascularization (CNV), diabetic macular edema (DME) and drusen are among the leading causes of irreversible vision impairment worldwide. Early detection and timely intervention are critical to preventing permanent visual loss. OCT has emerged as a non-invasive, high-resolution imaging modality that provides cross-sectional visualization of retinal microstructures, enabling detailed assessment of pathological changes. Despite its clinical value, manual interpretation of OCT scans is labor-intensive and subject to inter-observer variability. Consequently, automated deep learning–based systems have gained increasing attention as supportive tools for retinal disease diagnosis.

Among widely adopted Convolutional Neural Networks (CNN) architectures, ResNet [K. He et al., 2016] and DenseNet [G. Huang et al., 2017] represent two influential design paradigms for deep feature learning (**Figure 1**). ResNet introduces residual connections that allow identity mappings to bypass convolutional layers, effectively mitigating the vanishing gradient problem and stabilizing training. In contrast, DenseNet employs dense connectivity to maximize feature reuse and parameter efficiency. In the context of OCT, these architectures are particularly well-suited for extracting the subtle, multi-scale structural patterns characteristic of retinal pathologies.

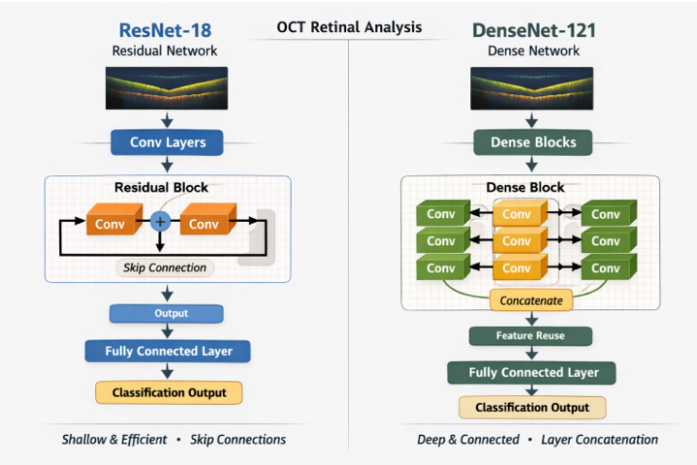

Figure 1: *Comparative CNN architecture of ResNet-18 and DenseNet-121 for OCT retinal image classification.*

A landmark study by Kermany et al. (2018) introduced a large-scale OCT dataset and showed that convolutional neural networks can achieve expert-level performance in distinguishing CNV, DME, Drusen, and Normal cases. Subsequent studies, such as Li et al. (2019) and Choudhary et al. (2023), further validated the effectiveness of transfer learning using pretrained convolutional neural network architectures (e.g., ResNet and DenseNet) for multi-class OCT retinal disease classification. These works demonstrated that deep feature extraction from ImageNet-pretrained models can achieve high diagnostic performance across CNV, DME, Drusen, and Normal categories. More recent research has explored advanced architectures, including hybrid CNN–Transformer models and attention-based mechanisms, reporting classification accuracies exceeding 95% on publicly available OCT datasets. These developments highlight the continued evolution of deep learning approaches toward improved feature representation, robustness, and clinical applicability. Moreover, while many studies emphasize overall accuracy, fewer explicitly analyze high-confidence misclassifications and probability calibration. Such factors are critical for safe deployment

in clinical decision-support systems. Therefore, this work focuses in part on optimizing and fine-tuning the model to reduce high-confidence misclassifications, thereby enhancing the credibility and reliability of the CNN-based diagnostic system.

2. Methodology

This study presents ResNet-18 and DenseNet-121 for OCT retinal disease classification under multiple training and optimization strategies to improve efficiency without sacrificing model transparency and reliability. Analysis focuses on (1) the impact of data augmentation on model generalization, (2) targeted fine-tuning on hard samples to address overconfident misclassifications, and (3) the use of Grad-CAM for interpretability. By evaluating metrics, robustness, calibration performance and visualization, this work aims to identify training strategies that not only maximize predictive performance but also improve the reliability and interpretability and clinical applicability in retinal analysis. The dataset consists of scans categorized into four classes: (a) CNV, (b) DME, (c) Drusen and (d) Normal (**Figure 2**). For this analysis we utilized the OCT2017 dataset [Kermany et al.], with images pre-split into standard training (60%), validation (20%) and test (20%) sets.

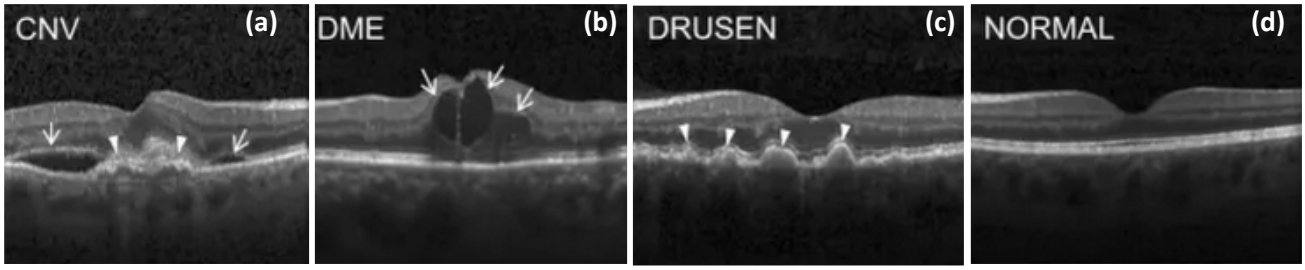

Figure 2: *(a) Choroidal Neovascularization, (b) Diabetic Macular Edema, (c) Drusen and (d) Normal [taken from Kumar et al. 2017]*

Transfer learning was implemented using pretrained models from the Torchvision library, leveraging ImageNet-1K weights to improve performance on medical imaging data. The original classification head was replaced with a linear layer mapping to four output nodes, corresponding to the diagnostic categories. Gradual unfreezing was conducted starting from the deepest (task-specific) layers and moving toward the shallowest (feature extraction) layers, eventually unfreezing the entire backbone. This approach allows the model to first adapt high-level semantic features while preserving general low-level representations. Specifically, the shallow layers capture general visual features such as edges and textures, which remain highly relevant in OCT imagery, whereas deeper layers adapt to disease-specific patterns, including fluid accumulation and retinal layer disruption. By preventing "catastrophic forgetting" we enhance the model's robustness and mitigate overfitting.

Augmentation techniques such as rotation, horizontal flipping, intensity variation, color jitter, Gaussian blur, resizing, normalization, and cropping were applied during training to improve model robustness and encourage the learning of invariant feature representations. These transformations simulate realistic acquisition variability and help prevent overfitting, while standard preprocessing transformations were consistently applied across all datasets to ensure uniform input preparation. Experiments were conducted on an NVIDIA A100 GPU using a class-weighted Cross Entropy Loss to mitigate class imbalance and were structured into 6 stages consisting of 15 epochs each. A cosine annealing scheduler was used, with both the scheduler and the learning rate reset at the beginning of each stage.

For DenseNet-121, batch size was initially set to 512, although due to memory constraints, was reduced to 256 from stage 4 onward. Learning rate was 1×10^{-4} for the first 3 stages and then reduced to 1×10^{-5}.

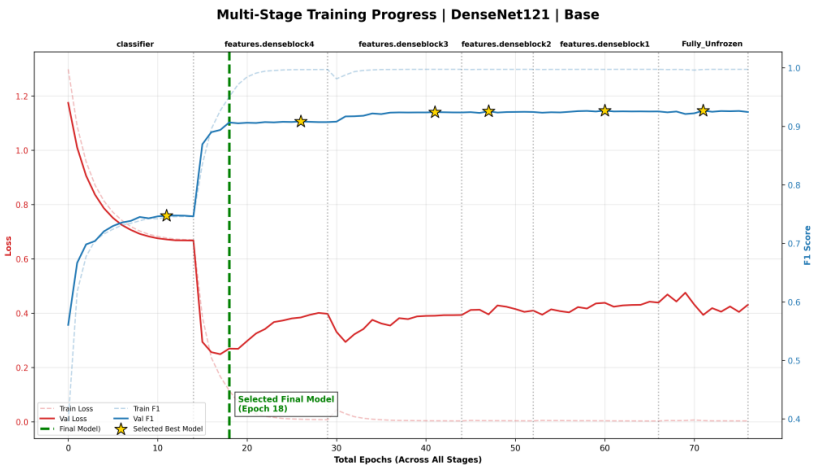

Figure 3*: Learning Curves for DenseNet121-Base. The plot shows Train - Val Loss and Train - Val F1-scores. Green dashed line indicates the model used for the metrics in Table 1. Stars indicate the best selected model during the unfreezing training process.*

For ResNet, batch size was consistently 512 and learning rate was resetting to 1×10^{-4}. During training, early stopping was applied but with different criteria: (a) ResNet: patience of 3 epochs (Δ_{min} = 0.0005) based on macro F1-Score. (b) DenseNet: patience of 5 for both macro F1-score (Δ_{min} = 0.0005) and validation loss, which was also checked to ensure that increase remains below 1.5 × threshold.

For the refinement phase, DenseNet-121-Aug underwent further training for 5 more epochs utilizing Stochastic Weighted Average (SWA). To address hard case misclassifications, a weighted random sampler was used, providing a 1.4 sampling boost for easy Drusen cases and 1.5 for all hard cases across all categories. Hard cases we identified as high-confidence misclassifications (conf > 0.9). Finally, Temperature scaling was applied to the refined model to ensure these confidence scores were well-calibrated for clinical reliability. Temperature scaling adjusts the softmax logits using a single learned parameter without altering the model's classification accuracy, thereby reducing overconfidence while preserving discriminative performance.

Similar to the ResNet18, training DenseNet-121 without augmentation typically led to overfitting upon unfreezing the last layers (**Figure 3**). This was characterized by a clear divergence between training (Train) and validation (Val) metrics (Loss and F1-Score), particularly within minority classes. In contrast, incorporating data augmentation significantly improved performance by increasing data variability and reducing overfitting. Implementation details, exploratory data analysis / training logs and supplementary data visualizations are available in our GitHub repository [GitHub].

3. Results

Both CNN architectures trained with data augmentation consistently outperform their non-augmented counterparts across all metrics in **Table 1**, including Accuracy, F1-Score and Expected Calibration Error (ECE). Data augmentation proved essential for OCT image analysis, as it enhanced dataset variability and reduced overfitting. Furthermore, the refined DenseNet-121-Aug with temperature scaling substantially reduced high-confidence misclassifications by **28.8%**. This refinement is particularly important in medical diagnostic applications, where erroneous high-confidence predictions may directly impact clinical decision-making.

Table 1: *Best model parameters based on without and with Augmentation and Refinement*

Model	**ResNet18**		**DenseNet121**		
Augmentation - Refinement	Base	Aug	Base	Aug	Aug Refined
Unfrozen layers	First 2	All	First 3	All	All
Accuracy	0.9348	0.9637	0. 9372	**0.9666**	0.9654
Precision	0.9016	0.9442	0.9097	0.9480	**0.9548**
Recall	0.9194	0.9539	0.9123	**0.9586**	0.9457
F1 score	0.9099	0.9489	0.9110	**0.9531**	0.9501
ECE	0.0085	**0.0080**	0.0117	0.0081	0.0104

Figure 4(a) illustrates the misclassification distribution by confidence level for the unrefined DenseNet-121-Aug. Data augmentation reduced the total errors, primarily below the 0.9 threshold. We defined 0.9 as the barrier for misclassifications since in medical applications, minimizing errors at high confidence levels is especially critical. Such predictions are more likely to directly influence clinical decision-making. The arrow and dotted box highlight the region of interest corresponding to misclassified OCT retinal images with Confidence greater than 0.9.

Furthermore, the percentage of high-confidence misclassifications is reported for each class. These high-confidence errors are of particular concern, as they indicate overconfident incorrect predictions. For example, in the case of CNV, 93 out of 7,432 samples (1.25%) are misclassified with confidence exceeding 0.9, as indicated by the red rectangle in **Figure 4(b)**. The total misclassifications above the threshold are 222 out of 16,699 samples. In **Figure 4(a),** the left inset displays the per-class Area-Under-Curve (AUC) values, where the lowest is observed for the Drusen class.

Hard case mining revealed 327 high-confidence misclassifications within the training set, primarily concentrated in the CNV ($n = 207$) and Drusen ($n = 49$) classes. Consistent with the errors identified in **Figure 3(b)**, these cases primarily involved confusion between CNV and Drusen. Implementing the class-specific refinement successfully addressed these 'confidently incorrect' samples, reducing the final test set high-confidence error count down to 158 from 222, representing the **28.8%**

reduction noted previously. This reduction confirms that the model became significantly more robust in the critical high-certainty region.

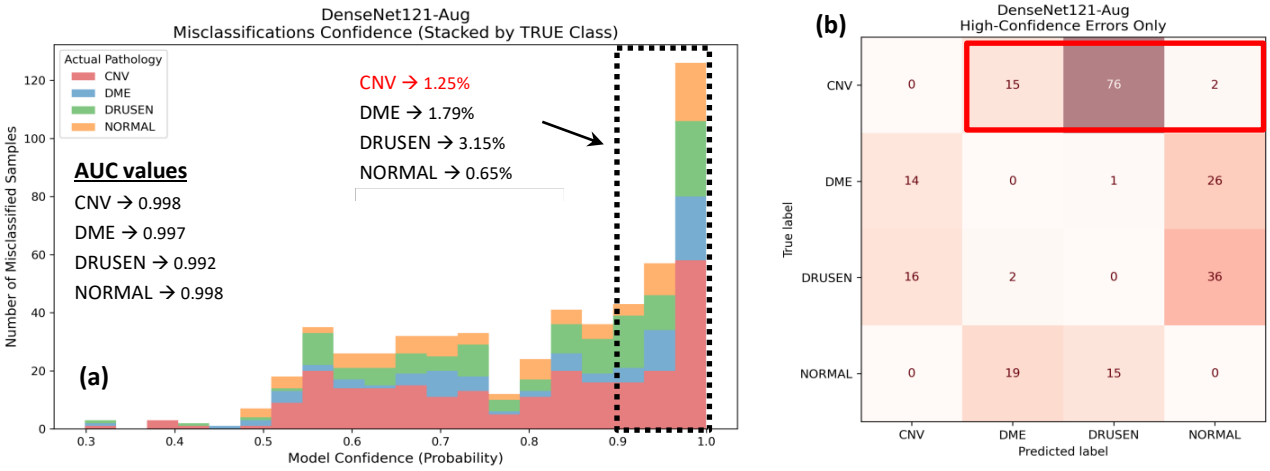

Figure 4: High-confidence error analysis for the unrefined DenseNet121-Aug. (a) Distribution of misclassifications across confidence scores; (b) Confusion matrix for high-confidence (Confidence > 0.9) test set errors.

Figure 5(a) presents the corresponding results for the refined DenseNet-121-Aug configuration. As illustrated, this strategy improves generalization performance, particularly by reducing overconfident errors in the CNV class. High-confidence misclassifications are substantially reduced, with total errors (confidence > 0.9) down to 158 out of 16,699 (0.94%). While Drusen remains the class with the highest proportion of high-confidence misclassifications, this redistribution of errors highlights a more effective classification boundary for the most clinically urgent pathology (CNV).

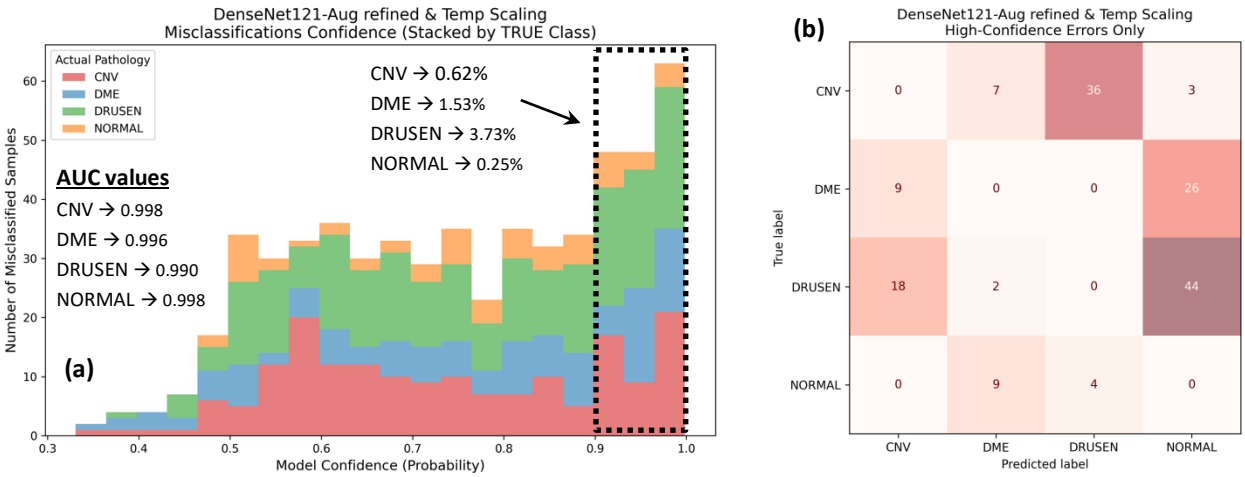

Figure 5: High-confidence error analysis for the refined DenseNet121-Aug. (a) Distribution of misclassifications across confidence scores; (b) Confusion matrix for high-confidence (Confidence > 0.9) test set errors.

To evaluate the clinical relevance of the model's decisions, Grad-CAM heatmaps were generated to visualize the retinal regions contributing most significantly to each prediction. By providing spatial explanations, Grad-CAM enhances model transparency and supports clinical interpretability. Such visualization tools can assist in identifying potential model biases or failure cases, further contributing to the development of safer and more reliable deep learning-based diagnostic systems.

Figure 6 presents representative Grad-CAM examples of correctly classified OCT images across all categories: (a) CNV, (b) DME, (c) Drusen, and (d) Normal for the augmented and refined DenseNet model. The heatmaps illustrate the class-specific activation regions, demonstrating that the network focuses on clinically relevant retinal structures associated with each pathology. These findings demonstrate that the refined configuration not only achieves high quantitative performance but also bases its predictions on anatomically meaningful regions, reinforcing its potential reliability for clinical decision-support applications.

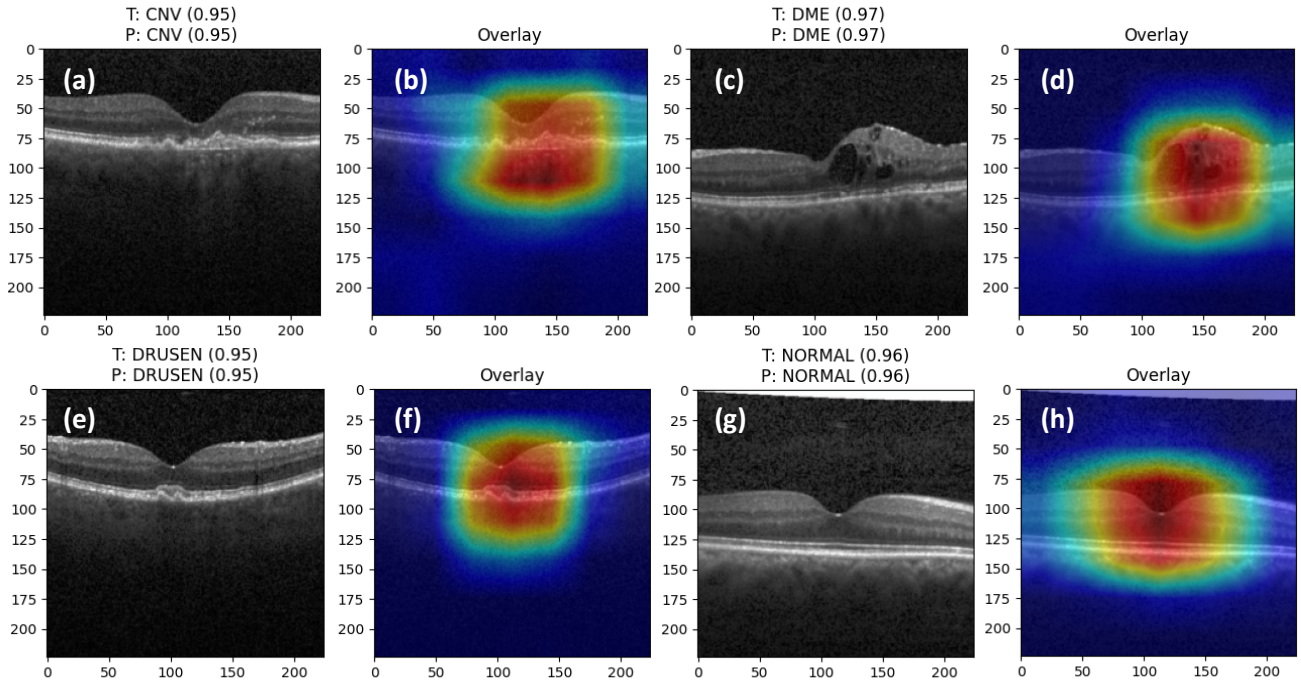

Figure 6: Representative OCT images corresponding to correct predictions of test set for (a) CNV, (c) DME, (e) Drusen, and (g) Normal obtained from the DenseNet121-Aug refined configuration. The corresponding Grad-CAM visualizations are shown in (b), (d), (f), and (h), respectively, overlaid on the original images to highlight the retinal regions contributing most strongly to each classification.

4. Conclusions

The results demonstrate that while data augmentation substantially improves model generalization, targeted fine-tuning through hard-case refinement enhances sensitivity for challenging classes. In particular, class-specific refinement reduced high-confidence misclassifications by 28.8%, contributing to more reliable predictions in clinically sensitive scenarios. Most notably, this approach led to an increase of Recall for CNV cases from 0.9666 to 0.9763 (**+0.97%**) ensuring higher detection for active pathological cases. These findings highlight the importance of addressing error behavior at high confidence levels, as deep neural networks remain prone to overconfident predictions under conditions of class imbalance.

Well-calibrated models provide confidence estimates that more faithfully reflect true predictive reliability, thereby supporting safer integration of automated systems into clinical workflows. Consequently, achieving both high discriminative performance and robust calibration is essential for the practical deployment of deep learning models in OCT-based retinal disease diagnosis. Future research should continue to explore strategies that jointly optimize accuracy, robustness, and interpretability to further enhance clinical trustworthiness. Future work will focus on addressing the data scarcity in underrepresented classes, such as Drusen which constitutes ~10,4% of the dataset. By exploring generative data augmentation techniques like GANs or Diffusion Models to synthesize realistic pathological features for these minority classes, we can provide the classifier with a more balanced training signal.

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